

MamaCare

Maternal Health Risk Prediction System

A data science project proposal for early pregnancy risk classification using basic maternal vital signs

Project area	Health data science / maternal health risk screening
Project type	Supervised classification with exploratory analysis, clustering, and neural network modelling
Proposed dataset	UCI Maternal Health Risk dataset
Target variable	RiskLevel: Low Risk, Mid Risk, or High Risk
Intended presentation output	Notebook, model comparison, visual report, and a small demonstration interface

1. Research Statement

This project investigates whether basic maternal health measurements can be used to classify pregnancy risk levels in a consistent and understandable way. The study will use routine clinical indicators such as blood pressure, blood sugar, body temperature, heart rate, and age to train predictive models that estimate whether a pregnancy record belongs to a low, mid, or high risk group. The work is intended as an academic decision-support prototype for learning and demonstration, not as a replacement for professional medical judgement.

2. Background of the Study

Maternal health remains a major public health concern because some pregnancy complications can worsen quickly when warning signs are missed or referral is delayed. Global health reports continue to show that many maternal deaths are linked to preventable or treatable complications such as bleeding, infection, high blood pressure disorders, and delivery-related complications. In practical clinical settings, especially low-resource facilities, health workers may need simple tools that help them review maternal vital signs quickly and identify mothers who may require closer attention.

The proposed project focuses on a narrow but realistic part of the problem: using already measurable vital signs to support early risk classification. The system will not diagnose a disease, prescribe treatment, or replace antenatal care guidelines. It will simply classify a record into a risk category and present the main measurements that contributed to the classification. This makes the project manageable for a final presentation while still addressing a meaningful health-sector problem.

3. Problem Statement

In many maternal health settings, risk signs such as abnormal blood pressure, high blood sugar, fever, and unusual heart rate may be recorded, but there may be no simple digital system that combines these measurements into an immediate risk category for review. This can make early triage less consistent, especially where facilities are busy or where community-level screening is being done with limited resources.

The problem addressed by this project is the lack of a simple, data-driven prototype that can use basic maternal measurements to identify likely low, mid, and high risk pregnancy records and explain the key factors behind the result. A well-designed model could support learning, awareness, and early referral discussions, provided that its limitations are clearly understood.

4. Aim of the Project

The aim of this project is to develop and evaluate a maternal health risk prediction system that classifies pregnancy records into low, mid, or high risk categories using basic maternal vital signs and data science techniques.

5. Specific Objectives

1. To clean and explore the maternal health risk dataset in order to understand the distribution of the variables and target classes.
2. To create useful derived features from existing vital signs, including pulse pressure, mean arterial pressure, and blood pressure risk indicators.
3. To apply clustering methods to identify natural groups of maternal health records before model training.
4. To train and compare several classification models, including simple baseline models and neural network models.
5. To tune the best model using validation results, regularization, early stopping, and appropriate performance metrics.
6. To interpret the final model using visualizations, confusion matrices, and feature-based explanations.
7. To prepare a small demonstration interface where a user can enter vital signs and receive a risk category for educational presentation purposes.

6. Research Questions

8. Which maternal vital signs show the strongest relationship with pregnancy risk level in the dataset?
9. Can engineered features such as pulse pressure and mean arterial pressure improve model performance?
10. How do simple machine learning models compare with a feedforward neural network on this classification task?
11. Which model gives the best balance between accuracy, class-wise recall, and interpretability?
12. How can the model output be explained in a way that a non-technical audience can understand during the final presentation?

7. Proposed Dataset

The proposed dataset is the Maternal Health Risk dataset from the UCI Machine Learning Repository. It contains 1,013 records, six input features, and one categorical target variable. The records were collected from maternal health-related settings and include common measurements that are easy to understand in a presentation. The target variable is RiskLevel, with three classes: Low Risk, Mid Risk, and High Risk.

Variable	Role	Description	Reason for use
Age	Input	Age of the pregnant mother	May be associated with pregnancy risk patterns
SystolicBP	Input	Upper blood pressure value	Important for detecting elevated blood pressure risk
DiastolicBP	Input	Lower blood pressure value	Complements systolic pressure in risk assessment
BS	Input	Blood sugar level	Useful for identifying glucose-related risk patterns
BodyTemp	Input	Body temperature	May indicate infection or abnormal body condition
HeartRate	Input	Heart rate in beats per minute	Shows cardiovascular strain or abnormal rate
RiskLevel	Target	Low Risk, Mid Risk, or High Risk	The class to be predicted by the model

A key limitation is that the dataset is not Kenyan hospital data. For that reason, the project will be presented as a learning prototype that demonstrates the method. If the system were to be used in practice, it would need to be retrained and validated using local clinical data and reviewed by qualified health professionals.

8. Proposed Methodology

The project will follow a standard data science workflow. The work will begin with data inspection and cleaning, then move to exploratory data analysis, feature engineering, model training, tuning, evaluation, and presentation through a small demo interface.

Stage	Work to be done	Expected output
Data loading	Load the dataset in Google Colab or Jupyter Notebook and inspect rows, columns, data types, duplicates, and target classes.	Initial understanding of the dataset
Data cleaning	Check for missing values, duplicates, unrealistic measurements, and inconsistent labels.	Clean dataset ready for analysis
EDA	Plot class distribution, histograms, boxplots, correlations, and risk-level comparisons.	Visual understanding of risk patterns
Feature engineering	Create pulse pressure, mean arterial pressure, blood-pressure flags, age bands, and glucose risk bands.	Improved modelling features
Scaling	Normalize numerical values using StandardScaler or a similar method.	Features suitable for neural network training
Clustering	Apply K-Means and hierarchical clustering to explore natural patient profiles.	Cluster plots and interpretation
Model training	Train baseline models and at least two neural network architectures.	Model comparison results
Tuning	Adjust layers, neurons, dropout, learning rate, batch size, and epochs.	Best-performing model selected
Evaluation	Use accuracy, macro F1-score, recall, precision, and confusion matrix.	Clear evidence of model performance
Demo	Build a simple form-based interface for entering maternal measurements.	Presentation-ready prototype

9. Model Development Plan

The modelling will not begin directly with a neural network. A simple baseline is necessary so that the group can show whether the more complex model actually improves performance. The proposed model comparison is as follows:

- Baseline model: predicts the most common risk class or uses a simple Logistic Regression model.
- Traditional models: Random Forest, Gradient Boosting, or Support Vector Machine for comparison.
- Shallow neural network: one hidden layer with ReLU activation and a softmax output layer.
- Deeper neural network: two or three hidden layers with dropout, batch normalization, early stopping, and tuned learning rate.

The final selected model will be the one that performs well on validation data while remaining understandable. If a simpler model performs almost the same as the deep model, that will be reported honestly. This is important because good data science is not about using the most complex model; it is about choosing the model that works reliably for the problem.

10. Evaluation Plan

The target variable has three classes, so the model should be evaluated beyond accuracy alone. Macro F1-score will be used as a major metric because it gives attention to each class, including the mid and high risk groups. Recall for the High Risk class will also be important because missing high-risk cases would be more serious than wrongly flagging a low-risk case for review.

Primary metric	Macro F1-score
Secondary metrics	Accuracy, precision, recall, confusion matrix, and class-wise performance
Validation method	Train-validation-test split or stratified cross-validation
Training visuals	Training loss, validation loss, training accuracy, and validation accuracy curves
Interpretation	Confusion matrix, feature influence, and explanation of common wrong predictions

11. Expected Deliverables

- A complete Jupyter or Google Colab notebook showing the full data science process.
- EDA charts and explanations of important risk patterns.
- K-Means and hierarchical clustering results with interpretation.
- At least two neural network architectures with training and validation plots.
- A model comparison table showing performance of all tested models.
- A simple demonstration interface for entering maternal vital signs and viewing the predicted risk level.
- A final report and presentation slides for NGAO Labs.

12. Scope of the Project

The scope of this project is limited to risk classification using the available dataset variables. The project will not attempt to diagnose specific pregnancy complications, prescribe treatment, or make automated clinical decisions. The output will be framed as a screening-support result that should be reviewed by a trained health worker.

13. Ethical and Practical Considerations

- The system should not collect names, phone numbers, or personal identifiers during the academic demo.
- Predictions should be presented carefully to avoid creating fear or false confidence.
- The interface should include a disclaimer that the result is not a medical diagnosis.
- A real deployment would require local clinical validation, data protection review, and medical supervision.
- The project should use respectful language when communicating risk levels.

14. Proposed Tools

Tool	Purpose
Google Colab / Jupyter Notebook	Development and execution of the data analysis workflow
Python	Main programming language
Pandas and NumPy	Data manipulation and numerical processing
Matplotlib and Seaborn	Data visualization
Scikit-learn	Preprocessing, clustering, baseline models, and evaluation metrics
TensorFlow/Keras	Building and training neural network models
Streamlit or Flask	Simple demonstration interface
GitHub	Version control and project sharing

15. Proposed Work Plan

Week	Main activities	Output
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Week 1	Confirm topic, download dataset, clean data, perform initial EDA.	Clean dataset and EDA charts
Week 2	Feature engineering, scaling, clustering, and cluster interpretation.	Cluster visualizations and notes
Week 3	Train baseline models and neural network models.	Model results and training curves
Week 4	Tune the best models, interpret results, and build the demo interface.	Improved model and working demo
Week 5	Finalize report, slides, testing, and presentation rehearsal.	Final presentation package

16. Expected Contribution

The expected contribution of this project is a clear demonstration of how data science can be used to support early maternal health risk screening using simple measurements. The project will help the group show practical skills in data preprocessing, visualization, clustering, supervised learning, neural network modelling, evaluation, and communication of results. It is also understandable to a non-technical panel because the problem is familiar and the input variables are common health measurements.

17. References

- UCI Machine Learning Repository. Maternal Health Risk dataset. Dataset page accessed July 2026. DOI: 10.24432/C5DP5D.
- World Health Organization. Maternal mortality fact sheet. Updated 7 April 2025.
- UNICEF Kenya. Transforming Maternal & Newborn Health: Kenya, UNICEF, and China Launch Lifesaving Initiative in Northeastern Kenya. Published 9 June 2026.